Naive Bayes classification

- Naive Bayes is a probabilistic classification algorithm.
- It is based on Baye's Theorem
- It predicts the class with the highest probability.
- Supervised ML algorithm used for classification.
- It assumes that all features are independent of each other (naive assumption)
- It is fast and efficient, both in training and prediction.
- It performs very well in text classification tasks (e.g., spam detection, sentiment analysis).
- It can handle categorical and numerical data (using different variants
- It suffers from the zero probability problem, which is solved using laplace smoothing.
- problem :- We want to classify whether a person will play cricket or not based on weather.

Outlook	Temperature	Play
Sunny	Hot	NO
Sunny	Hot	No
Overcast	Hot	yes
Rain	mild	yes
Rain	Cool	yes
Rain	Cool	No
overcast	mild	yes
Sunny	cool	yes

Total = 8 records

Predict $X = (\text{outlook} = \text{Sunny}, \text{Temperature} = \text{Cool})$

Step1: <u>Use Baye's Theorem</u>

$$P(C|X) = \frac{P(X|C) \cdot P(C)}{P(X)}$$

find ↳ given

we compare:

$P(\text{Yes}|X)$

$P(\text{No}|X)$

Step2: <u>Calculate Prior probabilities</u>

Yes = 5 , NO = 3

$P(\text{Yes}) = 5/8$

$P(\text{NO}) = 3/8$

Step3: <u>Likelihood for yes</u>

from dataset (only yes rows):

total yes = 5

- $P(\text{Sunny} \mid \text{Yes}) = 1/5$
- $P(\text{Cool} \mid \text{Yes}) = 2/5$

outlook and temperature are independent so that why we multiplied

$P(X \mid \text{Yes}) = (1/5) \times (2/5) = \boxed{2/25}$

↳ using Bayes' Theorem logic + independence (Naive assumption)

$P(\text{Sunny AND Cool} \mid \text{Yes}) = P(\text{Sunny} \mid \text{Yes}) \times P(\text{Cool} \mid \text{Yes})$

Step4: <u>Likehood for NO</u>

Outlook	Temp
Sunny	Hot
Sunny	Hot
Rain	Cool

Total NO = 3

$P(\text{Sunny} \mid \text{NO}) = 2/3$

$P(\text{Cool} \mid \text{NO}) = 1/3$

$P(X \mid \text{NO}) = (2/3) \times (1/3) = \boxed{2/9}$

Step 5: posterior probability

$$\boxed{P(C|X) \propto P(X|C) \times P(C)}$$

"proportional to"

for Yes

$$P(Yes|X) \propto P(X|Yes) \times P(Yes)$$

$$P(Yes|X) \propto (2/25) \times (5/8)$$

$$\frac{2 \times 5}{25 \times 8} = \boxed{0.05}$$

for NO

$$P(NO|X) \propto P(X|NO) \times P(NO)$$

$$\propto (2/9) \times (3/8)$$

$$= \frac{2 \times 3}{9 \times 8} = \boxed{0.083}$$

posterior probability is proportional to likelihood × prior.

Bayes formula

$$P(C|x) = \frac{P(x|C)\,P(C)}{P(X)}$$

$P(x)$ is same for all classes.

So, we ignore it when comparing classes

thats is why we write $\propto$ instead $=$

Step 6: Compare

Yes = 0.05

NO = 0.083

$\boxed{\text{NO is higher}}$

Output = NO (person will Not play cricket).

working:-

1. Start
2. Take input Dataset (features + class label)
3. Identify the target class (C) and input feature (X)

3.1 Compute prior probability for each class

4. Compute likelihood probability → P(X|C) for each class.
5. Apply Narve assumption.
   - assume all features are independent
   - multiply features probabilities
6. Use Bayes' Theorem

$$P(C|x) \propto P(X|C) \times P(C)$$

7. Calculate posterior probability for each class
   → for yes → for No (or other classes)
8. Compares all posterior probabilities
9. Select class with highest probability
10. Output final prediction → End.